SUMMARY

- Energy bands in solids are used to classify them into metals, insulators, and semiconductors
- In a N-type semiconductor, electrons are the majority charge carriers and holes are the minority carriers.
- In P-type semiconductor, holes are the majority charge carriers and electrons are the minority charge carriers.
- The thin region near the junction which is free from charge carriers (free electrons and holes) is called depletion region.
- When a PN junction diode is forward biased, the width of the depletion region decreases and the diode conducts.
- A PN junction diode in reverse biased condition functions as a open switch as it does not conduct. The width of the depletion region increases in this case.
- A forward biased PN junction diode functions as a rectifier. The process in which alternating voltage or alternating current is converted into direct voltage or direct current is known as rectification.
- The half wave rectifier rectifies one half of the input signal and produces a pulsating DC output.
- Full wave rectifier rectifies both halves of the input signal.
- Zener breakdown happens in a heavily doped PN junction diode when a strong electric field is applied.
- Avalanche breakdown occurs in lightly doped junctions which have wide depletion layers. It is due to the breaking of covalent bonds by the thermally generated minority charge carriers.
- Zener diode is a heavily doped PN junction diode which works in the reverse biased condition.
- Light emitting diode is a forward biased semiconductor device that emits emits visible or invisible light when energized. The recombination of minority charge carriers with the majority charge carriers in the respective regions release energy in the form of photons.
- A PN junction diode which converts an optical signal into electric signal is called a photodiode.
- When a photon of sufficient energy strikes the diode, it creates an electron-hole pair. These electrons and holes are swept across the p-n junction by the electric field created by reverse voltage before recombination takes place and in turn generates photo current.
- A solar cell, also known as photovoltaic cell, works on the principle of photovoltaic effect. Accordingly, the p-n junction of the solar cell generates emf when solar radiation falls on it. .
- A bipolar junction transistor (BJT) is a semiconductor device and there are two types in that: NPN and PNP.

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